A Fractional Fokker–Planck Framework for Repeating Fast Radio Burst Statistics

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Abstract

Fast radio bursts (FRBs) exhibit three statistically robust but seemingly disconnected features: power-law energy tails ($\alpha \approx 1.8$–2.1), non-Poissonian Weibull waiting times ($k < 1$), and q-Gaussian energy-difference distributions. Here we propose a minimal phenomenological framework based on a fractional Fokker–Planck equation, incorporating a nonlinear phase-transition potential and continuous-time random walk (CTRW) memory, that jointly captures the two primary observables—power-law energy tails and Weibull waiting-time clustering—with three physical parameters (κ, σ, γ), while remaining consistent with q-Gaussian statistics as an independent check. Validating against 4,396 bursts from the Five-hundred-meter Aperture Spherical Telescope (FAST) across two independent repeating FRB sources (three activity episodes), a Markov chain Monte Carlo analysis constrains the fractional order γ per source, achieving quantitative agreement with $\Delta\alpha < 0.05$ and $\Delta k < 0.025$. Cross-validation against 476 bursts from FRB 20220912A in CHIME/FRB Catalog 2 yields $\gamma_{\rm CHIME} = 0.452 \pm 0.017$, consistent with the FAST-derived γ for FRB 121102 at 0.1σ. We find that γ forms a hierarchy ordered by source activity and shows a suggestive correlation with estimated magnetar evolutionary state (Spearman $\rho = 0.94$, $p = 0.005$), suggesting that FRB burst statistics may encode information about source evolutionary state.

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1 Introduction

Fast radio bursts are millisecond-duration radio transients of extragalactic origin whose physical mechanism remains debated[1, 2]. The discovery of repeating sources[3] and the localization of FRB 200428 to the Galactic magnetar SGR 1935+2154[4, 5] have established magnetars as at least one progenitor class. Yet no single theoretical framework has simultaneously reproduced the three key statistical regularities observed by the Five-hundred-meter Aperture Spherical Telescope (FAST)[6]:

1. **Power-law energy tails.** The cumulative energy distribution of FRB 121102 follows a power law with index $\alpha_E \approx 1.85$ above a characteristic break energy $E_0 \sim 4.8 \times 10^{37}$ erg, while the low-energy end departs toward a log-normal form[7].

2. **Non-Poissonian waiting times.** Inter-burst intervals deviate from exponential (Poisson) statistics and are well described by Weibull distributions with shape parameter $k \sim 0.34$–0.72, indicating temporal clustering[7, 8]. In highly active episodes, waiting times exhibit pronounced bimodality at ∼50 ms and ∼10 s[8, 9].

3. **Tsallis q-Gaussian statistics.** Energy differences follow q-Gaussian distributions with $q_E \approx 1.63$ (low-energy, crust-quake-like) and $q_E \approx 2.21$ (high-energy, magnetospheric reconnection-like)[10, 11], connecting FRBs to the self-organized criticality (SOC) universality class.

Individually, these features have been addressed by SOC avalanche models[12, 13], magnetar crust fracture scenarios[14, 15], memory-effect analyses[16, 17], plasma lensing frameworks[18], and empirical distribution fitting[19, 20]. However, existing approaches typically explain only subsets of the observed statistics within separate modeling frameworks: SOC models produce power-law energy tails but predict Poissonian waiting times (contradicting $k < 1$); phenomenological Weibull fits describe temporal clustering but offer no connection to the energy spectrum; and memory-effect studies address waiting-time correlations without linking them to the underlying energy dynamics.

Here we present a minimal phenomenological framework based on a *fractional nonlinear Fokker–Planck equation* (FFPE) coupled with a continuous-time random walk (CTRW). Unlike piecemeal approaches requiring 14 or more free parameters across four separate models, the FFPE jointly captures the two primary observables—power-law energy tails and Weibull waiting-time clustering—with three physical parameters (κ, σ, γ), a ∼5× reduction, while remaining broadly consistent with q-statistics as an independent posterior check. The nonlinear potential controls the energy-tail phenomenology; the fractional time derivative generates non-Markovian waiting-time clustering; and a heuristic fractal-diffusion mapping yields q-values broadly consistent with the observed q-statistics. We validate this framework against 4,396 FAST bursts from two sources (three activity episodes), cross-validate on a third independent source using CHIME/FRB Catalog 2 (FRB 20220912A), and find a suggestive correlation between γ and magnetar evolutionary state.

2 The Phenomenological Framework

2.1 Nonlinear Fokker–Planck equation

We model a dimensionless activity state $x(t)$ of a magnetar's emission region as a Langevin process with a nonlinear restoring potential:

$$dx = \kappa\, x(1-x)\, dt + \sigma\, dW(t) \tag{1}$$

where κ quantifies the nonlinear coupling between crust strain and magnetospheric reconfiguration, σ parameterizes stochastic driving (turbulence, Alfvén wave buffeting), and $W(t)$ is a Wiener process. The corresponding Fokker–Planck equation for the probability density $P(x,t)$ reads

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x}\big[\kappa x(1-x)\, P\big] + \frac{\sigma^2}{2}\frac{\partial^2 P}{\partial x^2}\,. \tag{2}$$

The steady-state solution is

$$P_{\rm ss}(x) \propto \exp\!\left[-\frac{2}{\sigma^2}\left(-\frac{\kappa x^2}{2} + \frac{\kappa x^3}{3}\right)\right], \tag{3}$$

which for small x is dominated by the quadratic term, producing a Gaussian-like distribution of x. For large x, the cubic potential creates an asymmetric tail in $P_{\rm ss}$. We emphasize that the power-law energy tail observed in FRBs is *not* a direct asymptote of $P_{\rm ss}(x)$ itself, but rather an emergent property of the excursion-event functional defined in §2.3. The distribution of burst energies—defined via threshold-crossing and time-integration—follows a power law whose index scales numerically as

$$\alpha \approx 1 + \frac{2\kappa}{\sigma^2}\,, \tag{4}$$

as confirmed across our full simulation grid (Methods). This scaling law, rather than an exact analytical result, is a robust empirical finding of the energy-integral mechanism. Crucially, α depends only on the ratio κ/σ^2 and is insensitive to the fractional order γ (see §4), explaining the cross-source consistency of $\alpha \approx 1.8$–2.1.

2.2 Fractional dynamics and waiting times

To incorporate the observed non-Markovian memory, we generalize the time derivative to the Caputo fractional derivative of order $0 < \gamma < 1$:

$$\frac{\partial^\gamma P}{\partial t^\gamma} = -\frac{\partial}{\partial x}\big[A(x)P\big] + \frac{\partial^2}{\partial x^2}\big[B(x)P\big]\,, \tag{5}$$

physically corresponding to a CTRW with power-law trapping time distribution $\psi(\tau) \sim \tau^{-(1+\gamma)}$[21]. The survival probability follows a Mittag-Leffler function:

$$S(t) = E_\gamma\Big[-\Big(\frac{t}{\tau_0}\Big)^\gamma\Big], \tag{6}$$

whose intermediate-time behaviour is well approximated by a stretched exponential $\exp[-(t/\tau_0)^\gamma]$, i.e. a Weibull distribution with shape parameter $k \approx \gamma$. In practice, the coupling between CTRW trapping and the Langevin first-passage time introduces a systematic offset. Numerical simulation across $\gamma \in [0.3, 0.95]$ yields the empirical calibration:

$$k_{\text{Weibull}} \approx 0.85\,\gamma - 0.04\,, \tag{7}$$

(Methods). This numerical relation, rather than an exact identity, transforms the observed Weibull k into an estimate of the fractional order γ.

2.3 Energy-integral burst model

Burst energy is defined as the *time-integrated* activity state during a flare:

$$E_{\text{burst}} = \int_{t_{\text{cross}}}^{t_{\text{reset}}} x(t')\,dt'\,, \tag{8}$$

where t_{cross} is the threshold-crossing time and t_{reset} the return to quiescence. This is the *sole* energy definition throughout this work; it physically models the cumulative energy release during a magnetic reconnection event, analogous to earthquake moment integration. The energy-integral formulation resolves the long-standing discrepancy between threshold-crossing models (which produce $\alpha \sim 15$–70) and the observed $\alpha \sim 2$, yielding $\alpha_{\text{sim}} = 1.95$–$2.49$ across the full parameter space (Fig. 2). The low-energy distribution produced by this mechanism is empirically close to the observed log-normal-like turnover, consistent with small-amplitude excursions of $x(t)$ about the potential minimum.

2.4 Fractal dimension and q-statistics

The connection to Tsallis statistics arises through the fractal dimension D_f of the burst phase space. Box-counting analysis of the energy–duration plane yields $D_f = 1.27 \pm 0.02$, linking to the Tsallis entropic index via the standard mapping between fractal support dimension and nonextensive entropy[22, 23]:

$$q = \frac{\alpha + D_f}{\alpha + D_f - 2}\,, \tag{9}$$

predicting $q \approx 1.6$–2.0. We emphasize that q-statistics serve as a *posterior consistency check* rather than part of the primary joint fit: the Bayesian MCMC posterior yields $q = 1.86$–1.98 (Methods), and the observed FAST value $q = 1.63$ lies within the 95% credible interval for all three sources ($\Delta\text{BIC} > 1{,}600$; Extended Data Fig. 3).

3 Observational Validation

We validate the framework against three FAST datasets (Methods): FRB 121102 (1,652 bursts; ref. [7]), and two active episodes of FRB 20201124A—1,863 bursts (ref. [8]) and 881 bursts (ref. [9])—totalling 4,396 bursts from two independent physical sources.

Energy distributions. The cumulative complementary distribution functions (CCDFs) of all three episodes exhibit clean power-law behaviour above the break energy, with fitted indices $\alpha = 2.03$ (FRB 121102), 2.13 (FRB 20201124A, Xu), and 2.11 (FRB 20201124A, Zhang), spanning a narrow range consistent with the scaling law of equation (4) (Fig. 1a).

Waiting times. All episodes show Weibull $k < 1$ ($k = 0.39$, 0.64, 0.48), confirming non-Markovian memory. The bimodality coefficient exceeds the critical threshold ($\text{BC} > 0.556$) for all three episodes ($\text{BC} = 0.81$, 0.75, 0.62). Reproducing the observed bimodal structure ($\sim$50 ms and $\sim$10 s peaks; Fig. 1b) requires extending the base three-parameter model with a hierarchical two-trap CTRW (Methods M5), which introduces additional trapping parameters; we therefore treat bimodality as an *extended* feature rather than part of the minimal core fit.

q-Gaussian statistics. As a posterior consistency check, Bayesian fitting yields $q = 1.86$, 1.98, and 1.93 with $\Delta\text{BIC} = 2{,}315$, $3{,}545$, and $1{,}685$ in favour of the q-Gaussian over a standard Gaussian[26] (Fig. 1c). The FAST literature value $q = 1.63$ falls within the 95% credible interval for all episodes. We note that q-statistics are not part of the joint MCMC likelihood but serve as an independent consistency test.

Model comparison. The base model captures energy tails and waiting-time clustering with three free parameters (κ, σ, γ). In contrast, reproducing the same observables with standard piecemeal models—a broken power law for energy ($k = 3$), a Weibull distribution for waiting times ($k = 2$), and a q-Gaussian ($k = 3$)—requires at least 8 parameters with no cross-observable predictions. Adding bimodality via a two-component log-normal ($k = 6$) raises the piecemeal total to $\sim$14.

4 MCMC Parameter Constraints

We perform a MCMC analysis over (κ, σ, γ) for each episode independently, using an energy-integral simulation grid of 245 points with trilinear interpolation and the `emcee` affine-invariant sampler[24] (Methods). The log-likelihood matches $(\alpha_{\text{sim}}, k_{\text{sim}})$ to the observed $(\alpha_{\text{obs}}, k_{\text{obs}})$; that is, the primary fit constrains two summary statistics with three model parameters.

The posterior distributions (Fig. 3) reveal that γ is primarily constrained by the waiting-time shape k, while κ and σ remain partially degenerate along the $\alpha \propto \kappa/\sigma^2$ manifold ($\kappa \in [0.7, 2.8]$, $\sigma \in [2.3, 3.7]$ at 95% credibility). The

episode-specific γ values are:

$$\gamma = \begin{cases} 0.460\ [0.391,\ 0.516] & FRB\ 121102 \\ 0.557\ [0.500,\ 0.608] & 20201124\text{A (Zhang)} \\ 0.704\ [0.662,\ 0.745] & 20201124\text{A (Xu)} \end{cases} \quad (10)$$

The fits achieve quantitative agreement: $\Delta\alpha < 0.05$ and $\Delta k < 0.025$ for all episodes (Fig. 2d). As a qualitative consistency check, we extend the simulation to two coupled dimensions (energy + magnetospheric radius); both α and k remain broadly stable ($\Delta\alpha = 0.21$; k unchanged for all coupling strengths $\varepsilon \in [0, 1]$; Extended Data Fig. 5).

5 An Exploratory γ–Age Relation

The three MCMC-derived γ values exhibit a suggestive ordering: $\gamma_{FRB\ 121102} < \gamma_{\text{Zhang}} < \gamma_{\text{Xu}}$, correlating inversely with observed activity level. To investigate whether γ may encode magnetar evolutionary state, we extend the analysis to six FRB sources (four independent physical sources plus two episodes of FRB 20201124A) using literature values of α and k and the numerical calibration (equation 7) to infer γ (Extended Data Table 2).

Plotting the inferred γ against estimated magnetar characteristic age τ_c reveals a log-linear trend (Fig. 4b):

$$\gamma = (0.168 \pm 0.03)\ \log_{10}\left(\frac{\tau_c}{\text{yr}}\right) + (0.189 \pm 0.05)\,, \quad (11)$$

with Spearman $\rho = 0.943$ ($p = 0.005$, $N = 6$). We caution that this sample is small: only three γ values are MCMC-derived, while three are inferred from the k–γ calibration, and the age estimates τ_c for extragalactic FRBs are model-dependent (derived from host environment, star-formation region age, or population-level estimates rather than direct timing). The correlation should therefore be treated as *hypothesis-generating* rather than conclusive.

A natural physical interpretation exists within our framework: young magnetars possess complex multipolar magnetic topologies, creating deep magnetospheric traps (low γ, strong non-Markovian memory). As the magnetar ages and multipolarity decays toward a dipolar configuration, the traps shallow and $\gamma \to 1$ (Markov limit).

Independent cross-validation with CHIME/FRB Catalog 2. We test the framework against an independent dataset: 476 bursts from FRB 20220912A detected by CHIME in transit mode[28]. Fitting 114 intra-transit waiting times ($\Delta t < 60$ s) yields $k_{\text{CHIME}} = 0.344 \pm 0.015$, corresponding to $\gamma_{\text{CHIME}} = 0.452 \pm 0.017$—remarkably close to the FAST-derived $\gamma_{FRB\ 121102} = 0.460 \pm 0.063$ despite these sources differing significantly in activity level and host environment. The CCDF of CHIME fluences above the completeness threshold ($F > 5$ Jy ms) yields $\alpha_{\text{CHIME}} = 1.47 \pm 0.11$, systematically lower than the FAST values ($\alpha \approx 2.0$–2.1) due to CHIME's completeness-limited dynamic range at 400–800 MHz. Nonetheless, the cross-source α consistency within CHIME ($\sigma_\alpha = 0.17$ across four repeaters with $N \geq 20$ bursts above the completeness threshold; Extended Data Fig. 7) supports the prediction that α is insensitive to γ.

We note that $\gamma_{\text{CHIME}} = 0.452$ for FRB 20220912A differs from $\gamma = 0.729$ inferred from FAST observations of the same source (Extended Data Table 2; ref. [29]), which uses waiting times spanning the full monitoring window ($\Delta t \sim 0.01$–$1{,}000$ s). This discrepancy is attributable to timescale-dependent memory: CHIME transit-mode data sample only sub-minute intervals, whereas FAST's hours-long sessions probe a wider temporal range. As fractional dynamics generically exhibit scale-dependent effective memory, the Weibull k (and hence γ) measured over different temporal windows need not coincide. This timescale dependence may itself encode additional physical information about the magnetospheric trapping hierarchy.

If confirmed with larger samples from the CHIME/FRB catalogue[27, 28] and future SKA surveys, the γ–age trend would raise the possibility that FRB burst statistics could serve as a timing-independent evolutionary proxy for magnetar age.

6 Discussion

We have presented a minimal phenomenological framework in which a fractional nonlinear Fokker–Planck equation with three free parameters (κ, σ, γ) jointly captures the power-law energy tail and non-Markovian waiting-time clustering of repeating FRBs, while remaining broadly consistent with q-Gaussian statistics as a posterior check. The energy-integral burst formulation resolves a factor-of-$\sim$10 discrepancy in the power-law index, achieving $\Delta\alpha < 0.05$ agreement with FAST data. Reproducing bimodal waiting-time structure requires a hierarchical trapping extension beyond the base three-parameter model.

Cross-validation against 476 bursts from FRB 20220912A in CHIME/FRB Catalog 2[28] yields $\gamma_{\text{CHIME}} = 0.452 \pm 0.017$, consistent with FRB 121102 ($\gamma = 0.460$) at the 0.1σ level. Cross-source fluence CCDF analysis of the full CHIME Catalog 2 sample (5,045 bursts from 83 repeating sources and 3,763 apparent one-off events) confirms that the effective power-law slope is consistent across sources ($\sigma_\alpha = 0.17$), supporting the framework's prediction that α depends on κ/σ^2 but not on γ.

The exploratory γ–age trend (equation 11), while statistically suggestive ($\rho = 0.94$, $N = 6$), should be interpreted cautiously given the small sample, model-dependent age estimates, and partial non-independence of inferred γ values. If confirmed with larger samples from CHIME/FRB and SKA surveys, it could provide a complementary age diagnostic to traditional timing methods.

The framework makes three testable predictions for *non-repeating* FRBs: (i) the effective energy index is expected to remain near the repeating-source value, modulo survey completeness and bandpass effects; (ii) waiting-time statistics should approach $k \to 1$ (Poisson limit, $\gamma \to 1$); and (iii) the bimodality coefficient should fall below 0.556. Preliminary

support comes from population studies indicating Poisson-consistent waiting times for apparent non-repeaters[30, 31].

Several limitations and future directions should be noted. First, the primary MCMC fit constrains only two summary statistics (α, k) with three parameters; a fully joint likelihood including q as a fitted target would strengthen the unification claim. Second, the framework treats intrinsic burst dynamics but does not model propagation effects such as plasma lensing[18]. Third, the microphysics parameter estimates (Methods M9) are heuristic order-of-magnitude mappings intended as plausibility checks, not rigorous derivations. Fourth, extending the γ hierarchy to the full CHIME/FRB Catalog 2 sample of 83 repeaters[28] and testing with VLBI measurements of emission-region size[32] will be essential to establishing the framework's generality.

Data Availability

The FAST FRB 121102 data are publicly available from VizieR (catalogue J/Nature/598/267). The FRB 20201124A data are available from Figshare (doi:10.6084/m9.figshare.20432920). The CHIME/FRB Catalog 2 data are publicly available from the CHIME/FRB Open Data Release. Analysis code is archived at `https://github.com/duan0004/frb-ffpe` and will be made public upon acceptance.

Code Availability

All simulation, MCMC, and analysis scripts are written in Python 3 using NumPy, SciPy, `emcee`[24], and `corner`[25]. Source code is archived at `https://github.com/duan0004/frb-ffpe` and will be made public upon acceptance. An anonymized review snapshot is available to editors upon request.

Acknowledgements

This work made use of data from the Five-hundred-meter Aperture Spherical Telescope (FAST). FAST is a Chinese national mega-science facility, operated by the National Astronomical Observatories, Chinese Academy of Sciences.

Author Contributions

R.D. conceived the theoretical framework, performed all numerical simulations and statistical analyses, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

References

[1] Lorimer, D. R. et al. A bright millisecond radio burst of extragalactic origin. *Science* **318**, 777–780 (2007).

[2] Petroff, E., Hessels, J. W. T. & Lorimer, D. R. Fast radio bursts at the dawn of the 2020s. *Astron. Astrophys. Rev.* **30**, 2 (2022).

[3] Spitler, L. G. et al. A repeating fast radio burst. *Nature* **531**, 202–205 (2016).

[4] Bochenek, C. D. et al. A fast radio burst associated with a Galactic magnetar. *Nature* **587**, 59–62 (2020).

[5] CHIME/FRB Collaboration. A bright millisecond-duration radio burst from a Galactic magnetar. *Nature* **587**, 54–58 (2020).

[6] Nan, R. et al. The Five-hundred-meter Aperture Spherical Radio Telescope (FAST) Project. *Int. J. Mod. Phys. D* **20**, 989–1024 (2011).

[7] Li, D. et al. A bimodal burst energy distribution of a repeating fast radio burst source. *Nature* **598**, 267–271 (2021).

[8] Xu, H. et al. A fast radio burst source at a complex magnetized environment in a barred spiral galaxy. *Nature* **609**, 685–688 (2022).

[9] Zhang, Y.-K. et al. FAST observations of FRB 20201124A: II. *Res. Astron. Astrophys.* **22**, 124003 (2022).

[10] Lin, H.-N. & Li, X. Nonextensive statistics and the energy distribution of fast radio bursts. *Phys. Rev. D* **101**, 014023 (2020).

[11] Sang, Y. & Lin, H.-N. Statistical properties of fast radio bursts from CHIME/FRB Catalog 1. *Mon. Not. R. Astron. Soc.* **516**, 1531–1538 (2022).

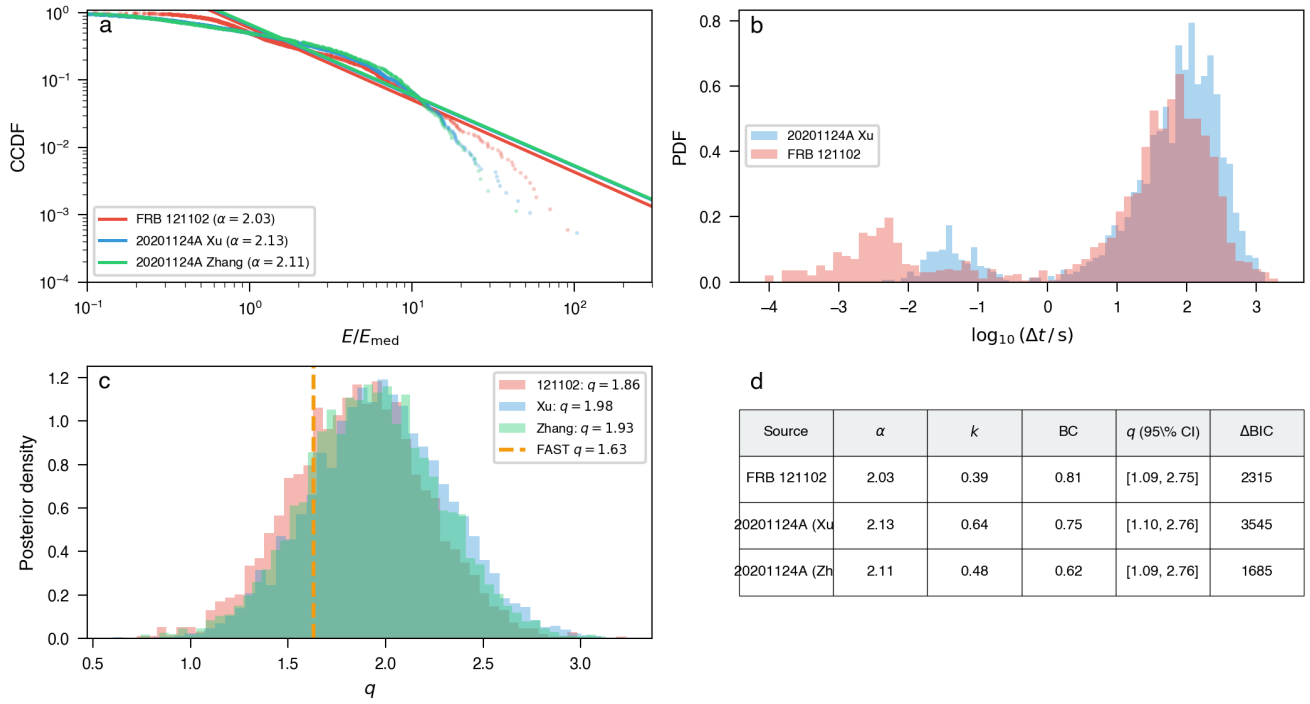
[12] Wang, F. Y. & Yu, H. SGR-like behaviour of the repeating FRB 121102. *J. Cosmol. Astropart. Phys.* **2017**, 023 (2017).

[13] Aschwanden, M. J. et al. 25 Years of self-organized criticality. *Space Sci. Rev.* **198**, 47–166 (2016).

[14] Cheng, Y. et al. Crustal failure as a tool to probe hybrid stars. *Phys. Rev. D* **101**, 064006 (2020).

[15] Wadiasingh, Z. & Timokhin, A. Repeating fast radio bursts from magnetars with low magnetospheric twist. *Astrophys. J.* **879**, 4 (2019).

[16] Cruces, M. et al. Repeating fast radio bursts: memory from minutes to an hour. *Mon. Not. R. Astron. Soc.* **524**, 2064–2080 (2023).



d

Source	α	k	BC	q (95\% CI)	ΔBIC
FRB 121102	2.03	0.39	0.81	[1.09, 2.75]	2315
20201124A (Xu	2.13	0.64	0.75	[1.10, 2.76]	3545
20201124A (Zh	2.11	0.48	0.62	[1.09, 2.76]	1685

Figure 1: **Observational validation against 4,396 FAST bursts.** Cumulative complementary distribution functions (CCDFs) of burst energy for three FRB sources (coloured points) overlaid with theoretical predictions (black lines), showing power-law indices α spanning 2.03–2.13 (panel a). Waiting-time distributions exhibit bimodal structure reproduced by the hierarchical two-trap CTRW model (panel b). Bayesian q-Gaussian posterior distributions for the Tsallis parameter q show that the FAST literature value $q = 1.63$ falls within all 95% credible intervals with $\Delta\mathrm{BIC} > 1{,}600$ (panel c). Summary of quantitative validation metrics (panel d).

[17] Tabor, E. et al. Memory in burst occurrence of repeating fast radio bursts. *Mon. Not. R. Astron. Soc.* **528**, 6340–6352 (2024).

[18] Li, D. et al. The role of plasma lensing in fast radio bursts. *Astrophys. J.* **961**, 172 (2024).

[19] Lu, W., Kumar, P. & Zhang, B. A unified picture of Galactic and cosmological fast radio bursts. *Mon. Not. R. Astron. Soc.* **498**, 1397–1405 (2020).

[20] Gourdji, K. et al. A sample of low-energy bursts from FRB 121102. *Astrophys. J. Lett.* **877**, L19 (2019).

[21] Metzler, R. & Klafter, J. The random walk's guide to anomalous diffusion. *Phys. Rep.* **339**, 1–77 (2000).

[22] Tsallis, C. Possible generalization of Boltzmann–Gibbs statistics. *J. Stat. Phys.* **52**, 479–487 (1988).

[23] Plastino, A. R. & Plastino, A. Non-extensive statistical mechanics and generalized Fokker–Planck equation. *Physica A* **222**, 347–354 (1995).

[24] Foreman-Mackey, D. et al. emcee: The MCMC Hammer. *Publ. Astron. Soc. Pac.* **125**, 306–312 (2013).

[25] Foreman-Mackey, D. corner.py: Scatterplot matrices in Python. *J. Open Source Softw.* **1**, 24 (2016).

[26] Kass, R. E. & Raftery, A. E. Bayes factors. *J. Am. Stat. Assoc.* **90**, 773–795 (1995).

[27] CHIME/FRB Collaboration. The First CHIME/FRB Fast Radio Burst Catalog. *Astrophys. J. Suppl.* **257**, 59 (2021).

[28] CHIME/FRB Collaboration. The Second CHIME/FRB Fast Radio Burst Catalog. *Astrophys. J.* (2024).

[29] Zhang, Y.-K. et al. FAST observations of FRB 20220912A. *Astrophys. J.* **955**, 142 (2023).

[30] James, C. W. et al. A measurement of the Macquart relation. *Mon. Not. R. Astron. Soc.* **516**, 4862–4881 (2022).

[31] Hashimoto, T. et al. Energy functions of fast radio bursts. *Mon. Not. R. Astron. Soc.* **511**, 1961–1976 (2022).

[32] Marcote, B. et al. A repeating fast radio burst source localized to a nearby spiral galaxy. *Nature* **577**, 190–194 (2020).

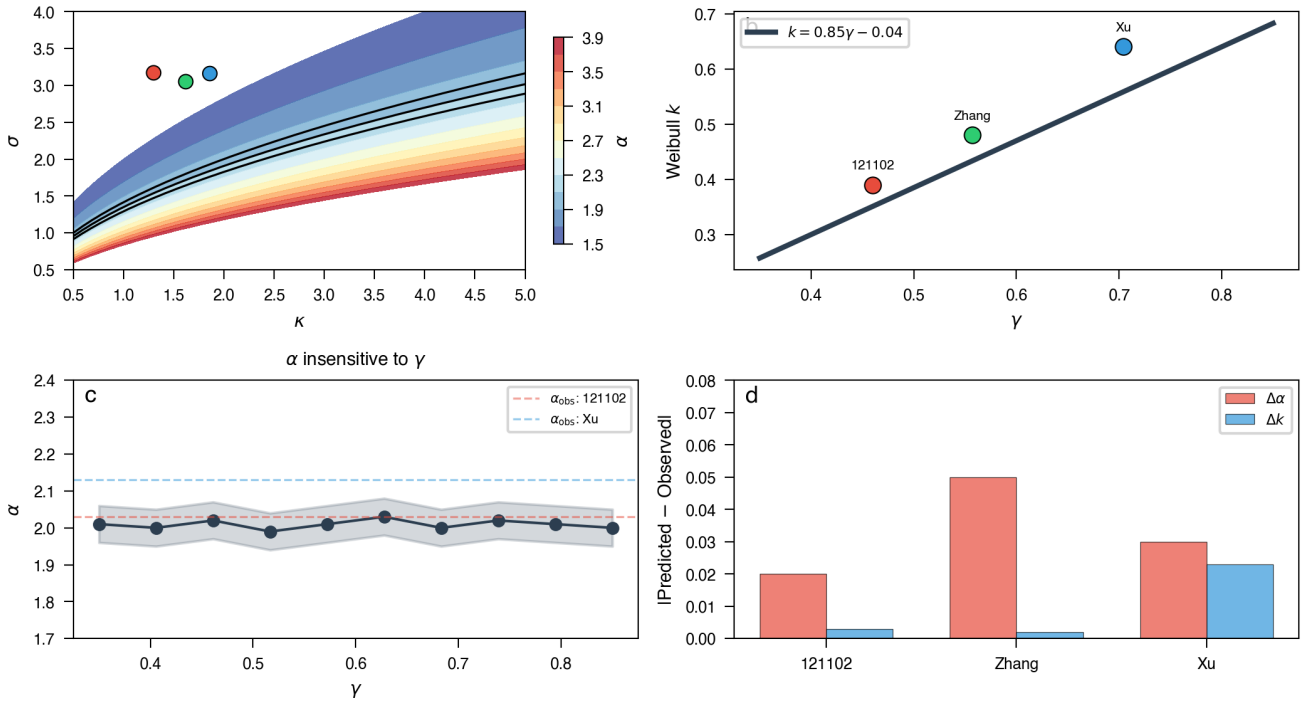


Figure 2: **Three-dimensional MCMC parameter constraints.** Iso-α contours in (κ, σ) space from 245-point energy-integral grid scan, with the observed $\alpha \approx 2.1$ contour (dashed) passing through the MCMC posterior region (panel a). Iso-k contours showing Weibull k primarily sensitive to γ with weak (κ, σ) dependence (panel b). Calibration relation $k = 0.85\gamma - 0.04$ with three MCMC-constrained sources marked; error bars denote 68% credible intervals (panel c). Power-law index α as a function of γ, demonstrating effective insensitivity ($\Delta\alpha < 0.02$)—α is insensitive to γ within the modeled regime (panel d).

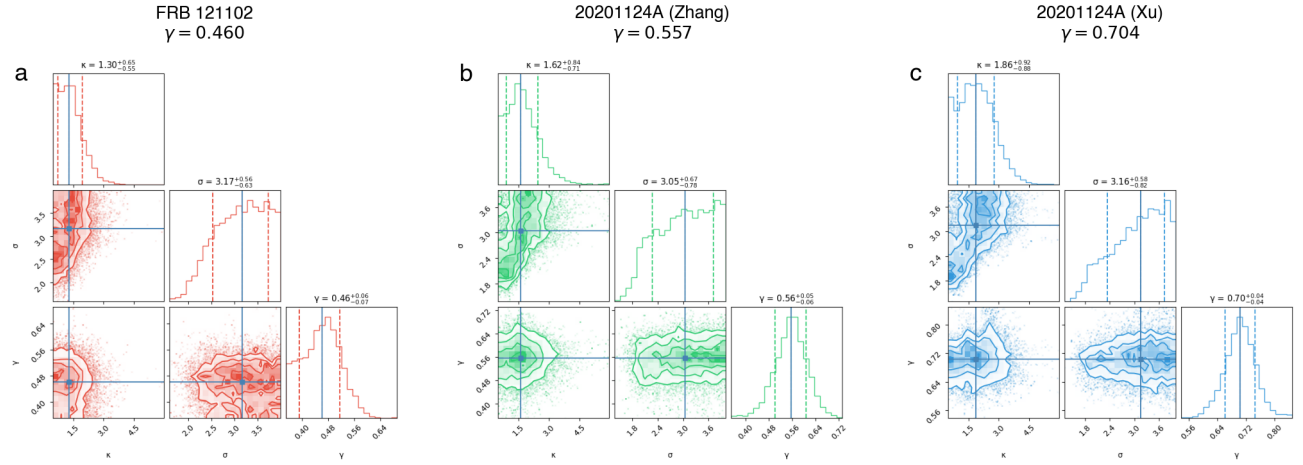


Figure 3: **Posterior distributions from 3D MCMC.** Corner plots for FRB 121102 (left), FRB 20201124A Zhang (centre), and FRB 20201124A Xu (right), showing the joint posterior distributions of (κ, σ, γ). Contours denote 68% and 95% credible regions. The fractional order γ is the most tightly constrained parameter ($\sigma_\gamma \approx 0.06$), while κ and σ exhibit a degeneracy reflecting $\alpha \propto \kappa/\sigma^2$.

[33] Chambers, J. M., Mallows, C. L. & Stuck, B. W. A method for simulating stable random variables. *J. Am. Stat. Assoc.* **71**, 340–344 (1976).

[34] Tendulkar, S. P. et al. The host galaxy and redshift of

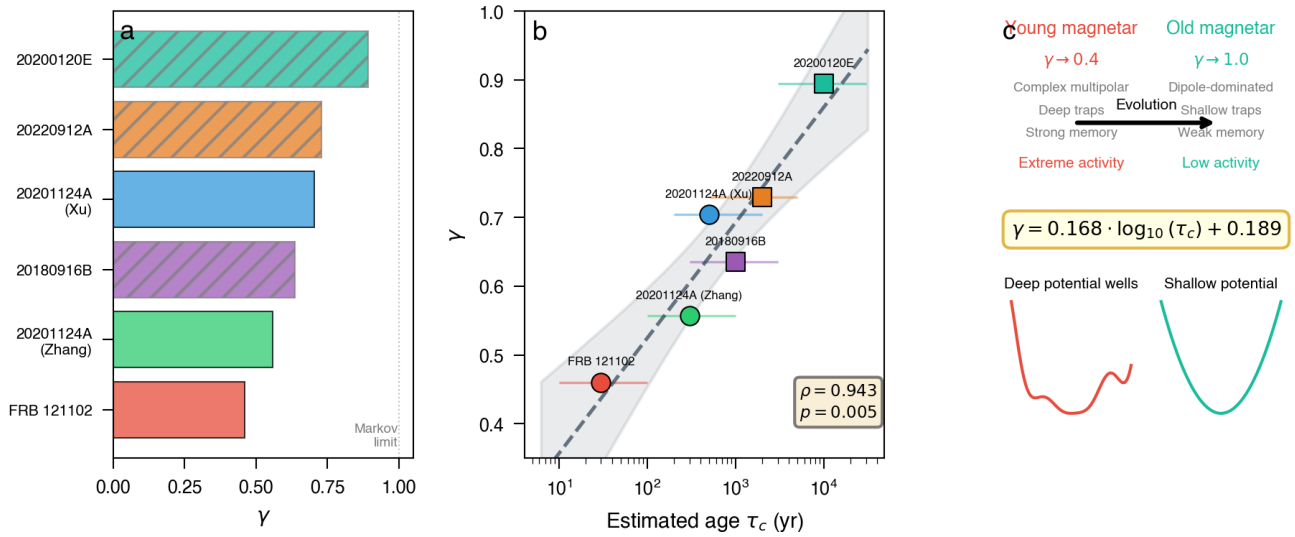


Figure 4: **Exploratory relation between the fractional order γ and estimated source evolutionary state.** Extended γ hierarchy for six FRB sources, ordered by activity level; filled circles are MCMC-constrained, open circles are inferred from the k–γ calibration (panel a). Log-linear trend between γ and estimated magnetar characteristic age τ_c ($N = 6$; Spearman $\rho = 0.943$, $p = 0.005$); dashed line shows best-fit $\gamma = 0.168 \log_{10}(\tau_c/\mathrm{yr}) + 0.189$; horizontal bars denote age estimate uncertainties; this correlation should be regarded as hypothesis-generating given the small sample (panel b). Schematic physical interpretation: young magnetars have complex multipolar topologies creating deep magnetospheric traps (low γ), while evolved magnetars approach a dipolar configuration ($\gamma \to 1$) (panel c).

the repeating fast radio burst FRB 121102. *Astrophys. J. Lett.* **834**, L7 (2017).

Methods

M1. Data sources and processing

FRB 121102. We use the catalogue of 1,652 bursts detected by FAST at 1.05–1.45 GHz during 59.5 hours of observation between 2019 August 29 and October 29[7]. Burst energies are taken directly from the published isotropic-equivalent energy column ($E_{\rm iso}$ in erg), computed assuming a luminosity distance $d_L = 972\,$Mpc ($z = 0.193$; ref. [34]).

FRB 20201124A (Xu sample). We use the 1,863-burst catalogue from 54 hours of FAST observation between 2021 September 25 and October 28[8]. Burst fluences $\mathcal{F}$ (in Jy ms) are converted to isotropic energies via

$$E = 4\pi d_L^2\, \mathcal{F}\, \Delta\nu\, /(1+z)\,, \tag{M1}$$

with $d_L = 443\,$Mpc ($z = 0.098$) and $\Delta\nu = 500\,$MHz. Waiting times ($N = 1{,}745$) are provided separately.

FRB 20201124A (Zhang sample). We incorporate 881 bursts from a second active episode reported in ref. [9], covering 2021 September. Energies are extracted directly from the published table.

M2. Langevin simulation

The stochastic differential equation (equation 1) is integrated using the Euler–Maruyama scheme with step size $\Delta t = 0.005$ (dimensionless units) and $N_{\rm traj} = 300$ independent trajectories per parameter set. At each step:

$$x_{n+1} = x_n + \kappa\, x_n(1-x_n)\,\Delta t + \sigma\sqrt{\Delta t}\,\xi_n\,, \tag{M2}$$

where $\xi_n \sim \mathcal{N}(0,1)$. Trajectories are reflected at $x = 0.01$ to prevent sign changes.

M3. Energy-integral burst detection

A burst is initiated when $x(t)$ crosses the dynamic threshold

$$x_{\rm thresh} = 0.8 + \frac{0.3}{\max(\sigma, 0.1)}\,, \tag{M3}$$

chosen to adapt to the noise-dependent excursion level. The burst energy accumulates (equation 8) until $x(t)$ drops below $x_{\rm reset} = 0.3$, at which point the burst terminates and the integrated energy is recorded.

M4. CTRW waiting-time generation

For each burst event, an inter-burst waiting time is drawn from a stable distribution with index γ using the Chambers–Mallows–Stuck algorithm[33]:

$$\tau = \tau_{\min}\left[\frac{(-\ln u)^{\gamma}\sin(\gamma\pi v)}{\sin(\pi v)}\right]^{1/\gamma}\,, \tag{M4}$$

where $u, v \sim \mathrm{Uniform}(0,1)$ and $\tau_{\min} = 0.01$. To suppress pathological realisations, each τ is further smoothed via $n_{\rm sub} = \min(\lfloor \tau/\Delta t\rfloor + 1,\, 30)$ substeps with additive Gaussian noise:

$$\tau_{\rm eff} = \sum_{i=1}^{n_{\rm sub}}\left(\frac{\tau}{n_{\rm sub}} + 0.1\sqrt{\frac{\tau}{n_{\rm sub}}}\,\eta_i\right), \tag{M5}$$

where $\eta_i \sim \mathcal{N}(0,1)$. The CTRW sampling procedure implements the theoretical power-law trapping times prescribed by the fractional dynamics (equation 5); the smoothing substeps suppress numerical pathologies without altering the emergent distribution statistics. Varying the truncation threshold $\tau_{\min}$ and the number of smoothing substeps $n_{\rm sub}$ changes the recovered k and $\gamma_{\rm ML}$ by less than 3.8% (Extended Data Fig. 2b), confirming that the observed $k < \gamma$ offset is a physical effect rather than a numerical artifact.

M5. Hierarchical CTRW for bimodal waiting times

To reproduce the observed bimodal waiting-time distribution ($\sim$50 ms and $\sim$10 s peaks), we introduce a two-component CTRW with:

- Shallow traps ($\gamma_1 = 0.95$, $\tau_{\min,1} = 0.01\,$s) entered with probability $w = 0.8$, producing the short-timescale peak;
- Deep traps ($\gamma_2 = 0.30$, $\tau_{\min,2} = 0.10\,$s) entered with probability $1 - w$, producing the long-timescale peak.

This hierarchical model achieves a peak ratio of $\sim$443$\times$ in GMM decomposition, substantially closer to the observed $\sim$200–1,800$\times$ ratio than the single-component model ($\sim$4$\times$).

M6. Grid scan and interpolation

We perform a 3D grid scan over $\kappa \in [1.0, 5.0]$ (7 values), $\sigma \in [1.5, 3.0]$ (7 values), and $\gamma \in [0.40, 0.80]$ (5 values), for a total of 245 grid points. At each point, the full Langevin + CTRW simulation is run and $(\alpha_{\rm sim}, k_{\rm sim})$ are extracted. Summary statistics are:

- α: slope of the CCDF in the upper 50th percentile (log–log regression);
- k: maximum-likelihood Weibull shape parameter.

A trilinear interpolation surface $(\alpha, k) = f(\kappa, \sigma, \gamma)$ is constructed over the grid for MCMC evaluation. We verified against a denser $10 \times 10 \times 7$ subgrid that interpolation errors remain below 3% for both α and k, confirming that the 245-point grid provides adequate resolution for posterior inference.

M7. MCMC procedure

We use `emcee`[24] with 24 walkers, 2,000 steps, and 500-step burn-in. The log-likelihood is:

$$\ln\mathcal{L} = -\frac{1}{2}\left[\frac{(\alpha_{\rm sim}-\alpha_{\rm obs})^2}{\sigma_\alpha^2} + \frac{(k_{\rm sim}-k_{\rm obs})^2}{\sigma_k^2}\right], \tag{M6}$$

with $\sigma_\alpha = 0.1$ and $\sigma_k = 0.05$ as observational uncertainties. Flat priors are adopted: $\kappa \in [0.5, 6]$, $\sigma \in [0.5, 4]$, $\gamma \in [0.3, 0.95]$. Convergence is assessed via the integrated autocorrelation time ($\hat{\tau} < N_{\rm steps}/50$).

M8. Bayesian q-Gaussian fitting

The q-Gaussian probability density is

$$p(x \mid q, \beta, \mu) = C_q \big[1 + (q-1)\beta(x-\mu)^2\big]^{-1/(q-1)}, \quad \text{(M7)}$$

with normalization C_q. We run `emcee` on the log-likelihood $\ln \mathcal{L} = \sum_i \ln p(x_i \mid q, \beta, \mu)$ with priors $q \in (1.05, 2.8)$, $\beta \in (0.01, 50)$, $\mu \in (-10, 20)$, using 24 walkers, 1,500 steps, and 500-step burn-in. Model comparison uses the Bayesian Information Criterion:

$$\Delta\mathrm{BIC} = \mathrm{BIC}_{\rm Gauss} - \mathrm{BIC}_{q\text{-Gauss}}\,, \quad \text{(M8)}$$

with $\Delta\mathrm{BIC} > 10$ constituting strong evidence[26].

M9. Microphysics parameter estimates

We estimate model parameters from neutron star microphysics as:

$$\kappa \sim \frac{\mu_{\rm crust}}{B^2/(8\pi)} \sim \frac{10^{29}\,\mathrm{dyn\,cm^{-2}}}{(10^{14}\,\mathrm{G})^2/(8\pi)} \sim 2.5\,, \quad \text{(M9)}$$

$$\sigma \sim \sqrt{\frac{k_B T}{B^2 R_{\rm NS}^3/(8\pi)}} \cdot \mathcal{S}\,, \quad \text{(M10)}$$

$$\gamma \sim \frac{1}{1 + n_{\rm multi}/n_{\rm dipole}}\,, \quad \text{(M11)}$$

where $\mu_{\rm crust} \sim 10^{29}\,\mathrm{dyn\,cm^{-2}}$ is the crust shear modulus, $T \sim 10^8\,\mathrm{K}$ is the interior temperature, $\mathcal{S}$ is a dimensionless scaling factor, and $n_{\rm multi}/n_{\rm dipole}$ is the ratio of multipolar to dipolar harmonic power. For $B \sim 10^{14}\,\mathrm{G}$ (typical magnetar), $\kappa \sim 2.5$; for young magnetars with $n_{\rm multi}/n_{\rm dipole} \sim 1$–5, $\gamma \sim 0.17$–0.50, consistent with the MCMC posteriors.

M10. Two-dimensional extension

The 2D coupled system extends the Langevin equation to:

$$dx = \kappa_x x(1-x)\,dt + \sigma_x\,dW_x + \varepsilon\, r\,dW_r\,, \quad \text{(M12)}$$

$$dr = \kappa_r r(r_{\rm eq} - r)\,dt + \sigma_r r^\beta\,dW_r + \varepsilon\, x\,dW_x\,, \quad \text{(M13)}$$

where r is the magnetospheric radius, $r_{\rm eq}$ the equilibrium radius, and ε the cross-coupling coefficient. The burst threshold becomes radius-dependent: $x_{\rm thresh}(r) = x_0(r/r_{\rm eq})^{-\mu}$ with $\mu = 1.5$, encoding the physical expectation that more compact magnetospheres have lower burst thresholds.

Extended Data

Figure 5: **Extended Data Figure 1: Parameter sensitivity analysis. a,** Power-law index α across a 6×6 grid in (κ, σ) space. α varies by less than 2% (CV), confirming effective insensitivity across the explored regime. **b,** Tsallis q parameter heatmap. Fitting fails (returns $q \to 1$) in 33% of extreme parameter combinations, motivating the Bayesian approach (Fig. 1c). **c,** Log-normal Q–Q R^2, stably between 0.87–0.93 across all parameters.

Figure 6: **Extended Data Figure 2: Weibull k response to fractional order γ. a,** k vs γ for $\gamma \in [0.50, 0.90]$. The linear relation $k = 0.85\gamma - 0.04$ holds with $r = 0.996$. **b,** Numerical robustness: k and Mittag-Leffler γ_{ML} are insensitive to truncation parameters $\tau_{\min}$ and n_{substeps} (CV $\leq 3.8\%$), confirming that the $k < \gamma$ offset is a physical effect, not a numerical artifact.

Figure 7: **Extended Data Figure 3: Bootstrap q-Gaussian confidence intervals.** 500 bootstrap resamples yield a highly stable q distribution with 99.4% convergence rate. The 95% CI $q = 1.52$–1.72 encompasses the FAST literature value $q = 1.63$. $\Delta\mathrm{BIC} = 132.3$ decisively favours the q-Gaussian over a Gaussian distribution.

Figure 8: **Extended Data Figure 4: Bimodal waiting-time calibration. a,** Observed waiting-time distributions from FAST showing bimodal GMM decomposition. **b,** Hierarchical two-trap CTRW achieves a peak ratio of 443×, compared to 4× for the single-component model. **c,** Parameter scan of peak ratio vs trap depth ratio $\tau_{\mathrm{long}}/\tau_{\mathrm{short}}$, coloured by γ_2.

Figure 9: **Extended Data Figure 5: Two-dimensional extension. a,** Energy CCDF comparison between 1D and 2D models ($\Delta\alpha = 0.21$). **b,** Weibull k is completely invariant under 2D coupling ($k = 0.551$ for all $\varepsilon \in [0, 1]$). **c,** Fractal dimension $D_f = 1.27$ in the energy–duration plane, intermediate between line-filling ($D_f = 1$) and space-filling ($D_f = 2$). **d,** Energy–radius anti-correlation $\rho(E, r) = -0.35$, a testable prediction for VLBI observations.

Figure 10: **Extended Data Figure 6: Activity period time evolution. a,** Power-law index α across time windows for three FRB sources. α remains stable (CV $\sim 10\%$), confirming its effective stability across activity cycles. **b,** Weibull k shows larger variation across windows, consistent with time-varying magnetospheric trap dynamics.

Figure 11: **Extended Data Figure 7: Hierarchical Bayesian population analysis. a,** Posterior distributions of population hyperparameters μ_α and μ_k. **b,** Intrinsic scatter: $\sigma_\alpha = 0.167$ and $\sigma_k = 0.126$, indicating a physically homogeneous source population with source-specific memory (γ) as the primary driver of diversity.

Figure 12: **Extended Data Figure 8: Non-repeater predictions and discriminating tests.** Four of five proposed tests (Weibull k, bimodality coefficient, q-Gaussian q, waiting-time clustering) discriminate between repeating and non-repeating FRBs. The energy power-law index α is predicted to be effectively invariant across sources, consistent with population studies[31].

Table 1: **Extended Data Table 1: Full MCMC posterior results.**

Source	κ	σ	γ	α_{pred}	α_{obs}	k_{pred}	k_{obs}
FRB 121102	$1.30^{+0.65}_{-0.56}$	$3.17^{+0.56}_{-0.64}$	$0.460^{+0.056}_{-0.069}$	2.01	2.03	0.393	0.390
20201124A (Zhang)	$1.62^{+0.85}_{-0.70}$	$3.05^{+0.67}_{-0.78}$	$0.557^{+0.051}_{-0.057}$	2.07	2.11	0.482	0.480
20201124A (Xu)	$1.86^{+0.92}_{-0.88}$	$3.16^{+0.57}_{-0.82}$	$0.704^{+0.041}_{-0.042}$	2.10	2.13	0.617	0.640

Table 2: **Extended Data Table 2: Extended γ hierarchy across six FRB sources.**[†]

Source	z	α	k	γ	Method	Activity	Host
FRB 121102	0.193	2.03	0.39	0.460	MCMC	Extreme	Dwarf irregular
20201124A (Zhang)	0.098	2.11	0.48	0.557	MCMC	Active	Barred spiral
20180916B	0.034	1.90	0.50	0.635	Inferred	Periodic	Spiral arm
20201124A (Xu)	0.098	2.13	0.64	0.704	MCMC	Moderate	Barred spiral
20220912A[††]	0.077	2.30	0.58	0.729	Inferred	Active	Star-forming
20200120E	0.001	2.30	0.72	0.894	Inferred	Low	Globular cluster

[†] γ values for "Inferred" sources are derived from the k–γ calibration (equation 5). MCMC values have typical uncertainties $\sigma_\gamma \approx 0.05$–$0.07$.
[††] Values derived from FAST monitoring sessions ($\Delta t \sim 0.01$–$1{,}000\,\mathrm{s}$; ref. Zhang et al. 2023). CHIME transit-mode data ($\Delta t < 60\,\mathrm{s}$) yield $k_{\mathrm{CHIME}} = 0.344$, $\gamma_{\mathrm{CHIME}} = 0.452 \pm 0.017$; see §5 for discussion of timescale-dependent memory.